156 RCs

1. Derivative

Chain Rule

If g is differentiable at x and f is differentiable at $g(x)$, then the composite function $F = f \circ g$ defined by $F(x) = f(g(x))$ is differentiable at x and F' is given by the product

$$F'(x) = f'(g(x)) \cdot g'(x)$$

In Leibniz notation, if $y = f(u)$ and $u = g(x)$ are both differentiable functions, then

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

The Power Rule Combined with the Chain Rule

If n is any real number and $u = g(x)$ is differentiable, then

$$\frac{d}{dx}(u^n) = nu^{n-1}\frac{du}{dx}$$

Alternatively,

$$\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1} \cdot g'(x)$$

Exercise 1

Find the derivative of the function:

1. $y = (4x - x^2)^{100}$
2. $y = 5^{-\frac{1}{x}}$
3. $y = e^{-2x} \cos 4x$
4. $y = (\frac{x^2+1}{x^2-1})^3$
5. $y = \dfrac{\arcsin x}{\arccos x}$
6. $y = [\sin (e^{(\sin x)^2})]^2$
7. $y = \arcsin \sqrt{\dfrac{1-x}{1+x}}$
8. $y = n^{n^x} + x^{n^n} + n^{x^n} \ (n > 0,\ n \neq 1)$

Exercise 1

1. $200(2-x)(4x-x^2)^{99}$
2. $ln5 \cdot 5^{-\frac{1}{x}} \cdot x^{-2}$
3. $-2e^{-2x}(cos4x + 2sin4x)$
4. $y' = 3(\frac{x^2+1}{x^2-1})^2(-\frac{2}{(x^2-1)^2}) \cdot 2x = -\frac{12x(x^2+1)^2}{(x^2-1)^4}$
5. $y' = \frac{(arcsinx)'arccosx - (arccosx)'arcsinx}{(arccosx)^2} = \frac{arccosx + arcsinx}{\sqrt{1-x^2}(arccosx)^2}$
6. $y' = 2[sin(e^{(sinx)^2})] \cdot cos(e^{(sinx)^2}) \cdot e^{(sinx)^2} \cdot 2sinxcosx = sin2x \cdot sin(2e^{(sinx)^2}) \cdot e^{(sinx)^2}$
7. $y' = (1-\frac{1-x}{1+x})^{-\frac{1}{2}} \cdot \frac{1}{2}(\frac{1-x}{1+x})^{-\dfrac{1}{2}}[-\frac{2}{(x+1)^2}] = -\frac{1}{\sqrt{2x(1-x)} \cdot (x+1)}$
8. $y' = n^{n^x+x}(lnn)^2 + n^n x^{n^n-1} + n^{x^n+1}x^{(n-1)}lnn$

Exercise 2

Find the second derivative of the following function:
(**Warning**: Don't forget to double check your first derivative!)

1. $y = \tan x$
2. $y = \dfrac{1}{x^3 + 1}$
3. $y = x \cos x$

Exercise 2

1. $y' = sec^2 x$

   $y'' = 2secx \cdot (tanx \cdot secx) = 2tanx \cdot sec^2 x$

2. $y' = -\dfrac{1}{(x^3+1)^2} \cdot 3x^2$

   $y'' = -3 \cdot \dfrac{2x(x^3+1)^2 - 2(x^3+1)3x^4}{(x^3+1)^4} = \dfrac{6x(2x^3-1)}{(x^3+1)^4}$

3. $y' = cosx - xsinx$

   $y'' = -2sinx - xcosx$

Exercise 3

Answer the following three questions based on $\dfrac{dy}{dx} = y'$:

1. Express $\dfrac{dx}{dy}$ with y'
2. Express $\dfrac{d^2x}{dy^2}$ with y' and y''
3. Express $\dfrac{d^3x}{dy^3}$ with y' ,y'' and y'''

Exercise 3

Solution:

1. $$\frac{dx}{dy} = \frac{1}{dy/dx} = \frac{1}{y'}$$
2. $$\frac{d^2x}{dy^2} = \frac{d}{dy} \cdot \frac{dx}{dy} = \frac{d}{dx}\frac{1}{y'}\frac{dx}{dy} = -\frac{y''}{(y')^3}$$
3. $$\frac{d^3x}{dy^3} = \frac{d}{dy}\frac{d^2x}{dy^2} = \frac{d}{dx}\frac{d^2x}{dy^2}\frac{dx}{dy} = -\frac{y'''(y')^3 - 3(y')^2(y'')^2}{(y')^6} \cdot \frac{1}{y'} = \frac{3(y'')^2 - y'y'''}{(y')^5}$$

Find y' if $\sin(x+y) = y^2 \cos x$

Differentiating implicitly with respect to x and remembering that y is a function of x, we get

$$\cos(x+y) \cdot (1+y') = y^2(-\sin x) + (\cos x)\,(2yy')$$

(Note that we have used the Chain Rule on the left side and the Product Rule and Chain Rule on the right side.) If we collect the terms that involve y', we get

$$\cos(x+y) + y^2 \sin x = (2y\cos x)y' - \cos(x+y) \cdot y'$$

So

$$y' = \frac{y^2 \sin x + \cos(x+y)}{2y\cos x - \cos(x+y)}$$

Exercise 7

Calculate the derivative of the following implicit function:

1. $y^2 - 2xy + 9 = 0$
2. $xy = e^{xy}$

Solution

1. $2yy' - 2y - 2xy' = 0, y' = \dfrac{y}{y-x}$
2. The derivative does not exist!

Exercise 4

Use **logarithmatic differentiation** to calculate the derivative of the following implicit function:

1. $y = (\frac{x}{x+1})^x$
2. $y = \sqrt[5]{\frac{x-5}{\sqrt[5]{x^2+2}}}$

Solution:

1. $lny = x[lnx - ln(x+1)]$
   $\frac{1}{y}y' = [lnx - ln(x+1)] + x[\frac{1}{x} - \frac{1}{x+1}]$
   $y' = (\frac{x}{x+1})^x[ln|\frac{x}{x+1}| + \frac{1}{x+1}]$
2. $lny = \frac{1}{5}ln|x-5| - \frac{1}{25}ln(x^2+2)$
   $y'\frac{1}{y} = \frac{1}{5(x-5)} - \frac{2x}{25(x^2+2)}$
   $y' = y[\frac{1}{5(x-5)} - \frac{2x}{25(x^2+2)}]$

Exercise 5

The *Bessel function* of order 0, $y = J(x)$, satisfies the differential equation $xy'' + y' + xy = 0$ for all values of x and its value at 0 is $J(0) = 1$.
(a) Find $J'(0)$.
(b) Use implicit differentiation to find $J''(0)$.

Exercise 5

Solution:

1. Take $x=0$, $0 + J'(0) + 0 = 0 \to J'(0) = 0$
2. $xy''' + 2y'' + y + xy' = 0$

   At $x = 0$, $2y'' + 1 = 0$, $y'' = -\dfrac{1}{2}$

Find y' if $y = f^{-1}(x)$

By definition, we get

$$x = f(y)$$

Apply $\frac{d}{dy}$ on both sides, we get

$$\frac{dx}{dy} = f'(y) = f'(f^{-1}(x))$$

On the other hand, $\frac{dy}{dx} = (f^{-1})'(x)$ According to $\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}}$, we reach the equation

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

Ex. 1

Find the derivative of the function $y = arcsin(x)$.

Ex. 2

Find the derivative of the function $y = arcsec(x)$.

Ex. 3

Find $(f^{-1})'(3)$ where $f(x) = x^3 + x + 1$.

More Exercises

10.

Find the derivative of the function $f(x) = (x\sin^2(x))^{\frac{1}{3}}$.

11.

The equation of the tangent to $f(x)$ at $x = 1$ is $y = -3(x-2)$, and knowing that $g(x) = f^{-1} + 3$, find the equation of the tangent to $g(x)$ at $x = 3$.

12.

Find the slope of the tangent to the curve $xy = \arctan(3y)$ when $y = \dfrac{1}{3}$.

Solutions:

1. Take logs on both sides: $lnf(x) = \frac{1}{3} \cdot (lnx + 2lnsinx)$
   therefore $f'(x) = \frac{1}{3}(xsin^2(x))^{\frac{1}{3}}(\frac{1}{x} + 2cot(x))$
2. $g(3) = f^{-1}(3) + 3 = 1 + 3 = 4$
   $g'(3) = [f^{-1}(x)]' = \frac{1}{f(f^{-1}(3))'} = \frac{1}{f'(1)} = -\frac{1}{3}$
   therefore $y - 4 = -\frac{1}{3}(x - 3)$
3. $x = \frac{arctan(3y)}{y}$
   apply $\frac{d}{dy}$ on both sides: $\frac{dx}{dy} = \frac{\frac{3y}{9y^2+1} - arctan(3y)}{y^2}$
   $k = \frac{dy}{dx} = \frac{4}{9(2-\pi)}$

Definition

$$f(x) \approx f(a) + f'(a)(x - a)$$

is called the **linear approximation** or **tangent line approximation** of f at a. The linear function whose graph is this tangent line, that is,

$$L(x) = f(a) + f'(a)(x - a)$$

is called the linearization of f at a.

Exercise 6

Use linear approximation to estimate $2.0006^{1.9998}$

Exercise 6

Solution:

$f(x) = 2^{x+2}$

$f'(x) = 2^{x+2} ln2$

$2^{1.9998} = f(0) + f'(0) \times (-0.0002) = 2^2 + 4ln2 \times (-0.0002) = 3.9994$

$g(x) = (2+x)^{1.9998}$

$g'(x) = 1.9998(2+x)^{0.9998}$

$2.0006^{1.9998} = g(0) + g'(0) \times 0.0006 = 2^{1.9998} + 1.9998 \times 2^{0.9998} \times 0.0006 = 4.0018$

Absolute Maximum and Absolute Minimum

Let c be a number in the domain D of a function f. Then $f(c)$ is the
- **absolute maximum** value of f on D if $f(c) \geqslant f(x)$ for all x in D.
- **absolute minimum** value of f on D if $f(c) \leqslant f(x)$ for all x in D.

Local Maximum and Local Minimum

The number $f(c)$ is a
- **local maximum** value of f if $f(c) \geqslant f(x)$ when x is near c.
- **local minimum** value of f if $f(c) \leqslant f(x)$ when x is near c.

Critical number

A **critical number** of a function f is a number c in the domain such that either $f'(c) = 0$ or $f'(c)$ does not exist, or the end points of a region if it is bounded.

local maximum/minimum $\rightarrow$ critical number

Related Theorems

The Extreme Value Theorem

If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers c and d in $[a, b]$.

Fermat's Theorem

If f has a local maximum or minimum at c, and if $f'(c)$ exists, then $f'(c) = 0$.

Rolle's Theorem

Let f be a function that satisfies the following three hypotheses:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b).
3. $f(a) = f(b)$

Then there is a number c in (a, b) such that $f'(c) = 0$.

Lagrange Mean Value Theorem

Let f be a function that satisfies the following hypotheses:
1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b).

Then there is a number c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

or, equivalently, $f(b) - f(a) = f'(c)(b - a)$

Cauchy Mean Value Theorem (Extended Mean Value Theorem)

Let f, g be two functions that satisfy the following hypotheses:
1. f, g is continuous on the closed interval $[a, b]$.
2. f, g is differentiable on the open interval (a, b).
3.$x \in (a, b), g'(x) \neq 0$
Then there is a number c in (a, b) such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$$

or, equivalently, $(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c)$

Exercise 7

Verify that the function satisfies the hypotheses of the Mean Value Theorem on the given interval. Then find all numbers c that satisfy the conclusion of the Mean Value Theorem.

1. $f(x) = x^3 - 3x + 2, \quad [-2, 2]$
2. $f(x) = \ln x, \quad [1, 4]$

Exercise 7

Solutions:

1. f(-2) $= 0$, f(2) $= 4$

   $$k = \frac{f(2) - f(-2)}{4} = 1$$

   $$f'(x) = 3x^2 - 3, 3c^2 - 3 = 1$$

   $$c = \frac{2\sqrt{3}}{3}$$

2. $$k = \frac{f(4) - f(1)}{3} = \frac{2ln2}{3}$$

   $$f'(x) = \frac{1}{x}$$

   $$d = \frac{3}{2ln2}$$

Relationship between f' and f

Increasing/Decreasing Test

If $f'(x) > 0$ on an interval, then f is increasing on that interval.
If $f'(x) < 0$ on an interval, then f is decreasing on that interval.

The First Derivative Test

Suppose that c is a critical number of a continuous function f.
If f' changes from positive to negative at c, f has a local maximum at c.
If f' changes from negative to positive at c, f has a local minimum at c.
If f' does not change sign at c, f has no local maximum or minimum at c.

Relationship between f'' and f

Concave upward/downward

If the graph of f lies above all of its tangents on an interval I, then it is called **concave upward** on I. If the graph of f lies below all of its tangents on an interval I, then it is called **concave downward** on I.

Concavity Test

If $f''(x) > 0$ for all x in I, then the graph of I is concave upward on I.
If $f''(x) < 0$ for all x in I, then the graph of I is concave downward on I.

Relationship between f'' and f

Inflection point

A point P on a curve $y = f(x)$ is called an **inflection point** if f is continuous there and the curve changes from concave upward to concave downward or from concave downward to concave upward at P.

The Second Derivative Test

Suppose f'' is continuous near c.
If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c.
If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c.

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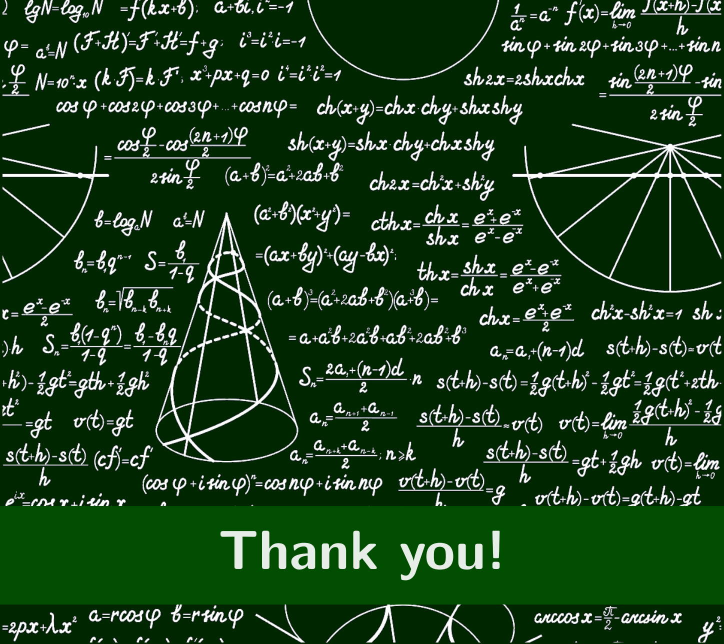

Thank you!